



Bharat Institute of Engineering and Technology

DawaRaksh

IoT-Enabled Smart Container for Monitoring Temperature-Sensitive Drugs and Vaccines

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Temperature-sensitive drugs and vaccines can lose effectiveness when exposed to temperature excursions, humidity variation, leakage, mishandling or unauthorized opening during storage and transit.

Operational Context

- Sensitive medical products require stable handling conditions
- Manual inspection is periodic and cannot guarantee continuous visibility.
- A monitored container provides event-level awareness at the point of storage.

Why This Matters

- Temperature drift can reduce product reliability.
- Leaks or wet contact may damage labels, packaging and contents.
- Door opening and stock depletion should be captured as measurable events.

Project Position

- The objective is monitoring, logging and alerting
- The design is modular

Existing low-cost solutions typically observe only one or two factors, such as temperature or quantity, and fail to provide an integrated view of internal condition, stock state, access state, wet contact and fault-aware communication backup.

Observed Limitations

- Single-parameter systems do not reflect the complete state of a sensitive storage container.
- Cloud-only systems are vulnerable during connectivity loss.

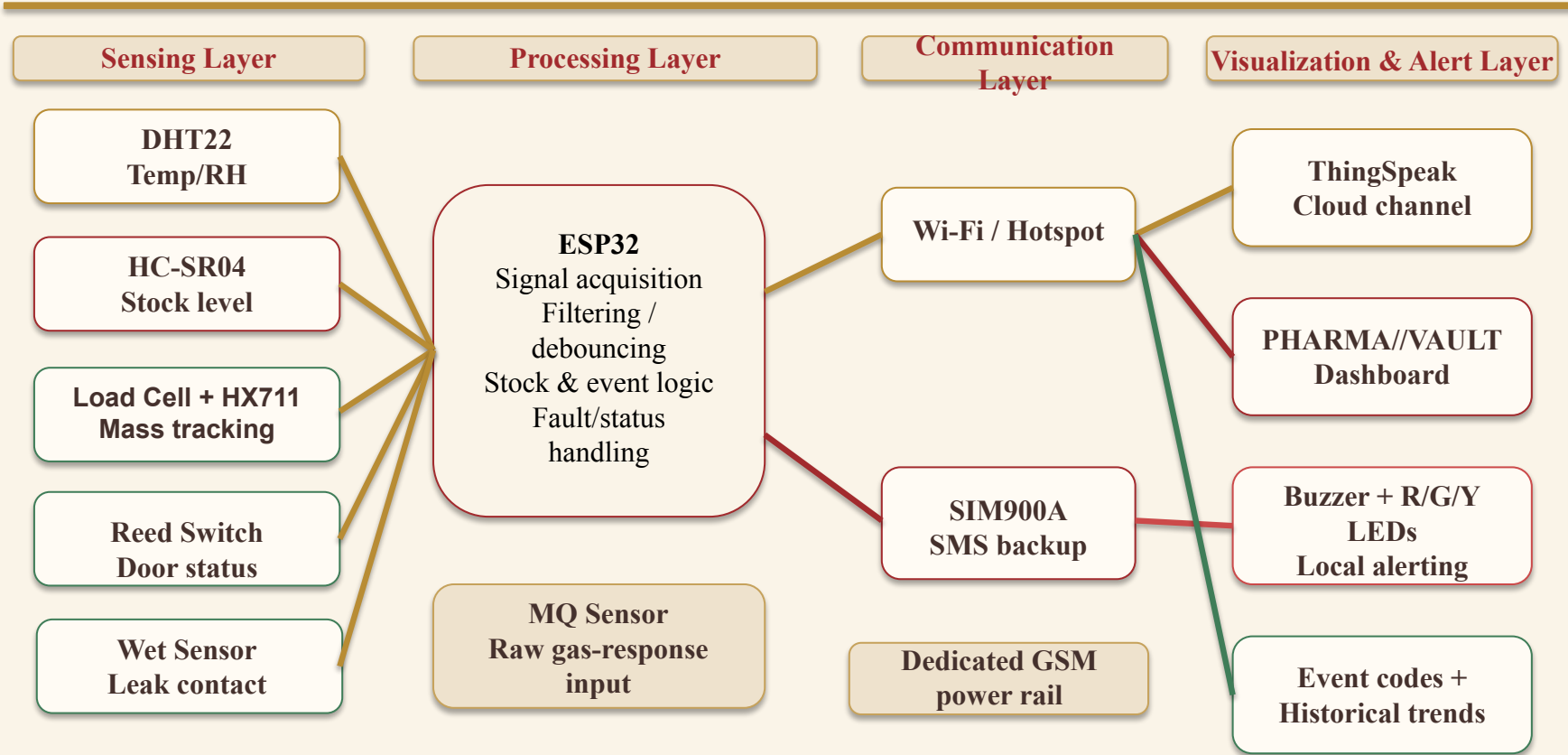
Refined Problem Definition

- Develop a realistic sensor-centric monitoring prototype around ESP32.
- Track temperature, humidity, stock change, access opening, leakage and anomaly indications.
- Provide local alerts, cloud visibility and GSM backup in a single architecture.

Year	Title / Venue	Method / Technology	Key finding	Research gap identified
2023	IoT-driven Smart Packaging for Pharmaceuticals (IEEE)	Environmental and handling sensors in smart packaging	Demonstrates the value of integrated sensing for product integrity and patient safety	Packaging concept only; does not define a student-feasible container prototype with staged implementation
2023	Intelligent Medicine Box (E3S Web of Conferences)	Arduino-based box automation for reminders and access control	Useful for box-level interaction and user access ideas	Reminder-oriented; limited focus on multi-event environmental monitoring
2022	IoT Based Blockchain Temperature Monitoring and Fake Medicine Prevention (MEJAST)	IoT with blockchain for traceability and temperature logging	Shows importance of trustworthy cold-chain data	Strong on traceability, but weak on physical sensing, leak detection and local alert redundancy
2024	Automatic Environmental Monitoring for Medicine Storage (Vol. 4, No. 1)	DHT22 and air-quality sensing with IoT monitoring	Validates medicine-storage monitoring using temperature and humidity sensing	Focuses mainly on room-level environment rather than container-level stock, access and wet events

Capability	Smart medicine box	Storage room monitor	Traceability system	Proposed DawaRaksh
Temperature monitoring	Partial	Yes	Yes	Yes
Humidity monitoring	Limited	Yes	Limited	Yes
Stock/fill estimation	Occasional	No	No	Yes
Mass change detection	Rare	No	No	Yes
Door/opening event	Some versions	No	No	Yes
Leak / wet detection	Rare	No	No	Yes
Gas anomaly input	Rare	Air-quality only	No	Yes (qualitative raw)
Cloud history	App-centric	Yes	Yes	Yes
Backup SMS alerting	Usually absent	Usually absent	Usually absent	Yes

- 1 Monitor internal temperature and relative humidity using DHT22.
- 2 Estimate stock/fill level using an HC-SR04 ultrasonic sensor.
- 3 Measure mass and detect significant mass change using a load cell and HX711.
- 4 Detect door-opening incidents using a magnetic reed switch.
- 5 Capture qualitative raw gas-response changes using an MQ-series sensor.
- 6 Upload real-time data to ThingSpeak through ESP32 Wi-Fi communication.
- 7 Provide local buzzer/LED alerts and backup SMS notification through SIM900A.
- 8 Visualize current and historical values on the PHARMA//VAULT dashboard.



Development Environment

- Arduino IDE 2.0 for coding, compiling and uploading ESP32 firmware
- ESP32 board support package by Espressif Systems
- Circuit Designer or equivalent for wiring and circuit planning
- Serial monitor for bench debugging and module bring-up

Firmware / Library Stack

- Embedded C/C++ running on ESP32
- Libraries: WiFi.h, ThingSpeak, DHT Sensor Library, Adafruit Unified Sensor, HX711 library
- Logic: thresholding, filtering, debouncing, event-code generation and timed upload scheduling
- UART AT-command handling for SIM900A SMS backup

Cloud & UI Layer

- ThingSpeak private channel for time-series data logging
- PHARMA//VAULT dashboard using HTML, CSS and JavaScript
- Live and historical visualization using channel read access
- Credentials isolated in a secrets file

Hardware block	Component	Technical purpose
Controller	ESP32 DevKit V1	Main processing, Wi-Fi connectivity, event logic and UART communication
Environmental sensing	DHT22 / AM2302	Temperature and relative humidity measurement
Stock sensing	HC-SR04	Distance-based fill-level / stock estimation
Mass sensing	5 kg load cell + HX711	Mass measurement and baseline-change detection
Access & leak sensing	MC-38 reed switch + wet sensor	Door-open and liquid-contact event detection
Gas-response sensing	MQ-series sensor	Qualitative raw gas anomaly indication
Alerts & comms	Buzzer, R/G/Y LEDs, SIM900A	Local indication and backup SMS notification
Physical build	Insulated box / thermocol enclosure / safe sensor mounting	Prototype housing for monitored medicine-container concept

Prototype Outcomes

- Continuous sensing of temperature, humidity, stock level, mass and container incidents
- Event-based indication of abnormal conditions such as door opening, leakage, stock drop or gas anomaly
- A realistic monitored-container demonstrator suitable for academic presentation and testing

Communication Outcomes

- Real-time uploads to ThingSpeak with historical records
- PHARMA//VAULT dashboard for intuitive status visualization
- Backup SMS delivery for critical alerts when configured

Academic Outcomes

- A stronger and more realistic major-project scope
- Documented architecture, wiring, firmware structure and validation roadmap
- Foundation for future calibration, enclosure refinement and performance testing

DawaRaksh has been refined into a technically feasible, monitoring-first IoT prototype for temperature-sensitive pharmaceutical containers, combining multi-sensor input, ESP32-based decision logic, cloud visualization, local alerts and GSM backup.

Immediate Next Step

- Begin Phase-1 bench test: ESP32 + HC-SR04 + hotspot + ThingSpeak.
- Validate the first live upload, then integrate additional sensors one by one.
- Freeze the final BOM and proceed toward full prototype construction.

ESP32

Unified
controller
architecture

3

Alert layers:
local, cloud,
GSM

6+

Sensor inputs
and states

- [1] R. Raman et al., “IoT-driven Smart Packaging for Pharmaceuticals: Ensuring Product Integrity and Patient Safety,” IEEE, 2023.
- [2] M. Shalini et al., “Intelligent Medicine Box,” E3S Web of Conferences, Vol. 399, 2023.
- [3] M. Jammula, “IoT Based Blockchain Temperature Monitoring and Fake Medicine Prevention,” Middle East Journal of Applied Science & Technology, 2022.
- [4] R.K. Harahap et al., “IoT-based Automatic Environmental Monitoring and Regulation System for Medicine Storage with Integration of Temperature, Humidity and Air Quality Sensors,” Vol. 4, No. 1, Dec. 2024.
- [5] ESP32, DHT22, HC-SR04, HX711 and SIM900A datasheets / library documentation.